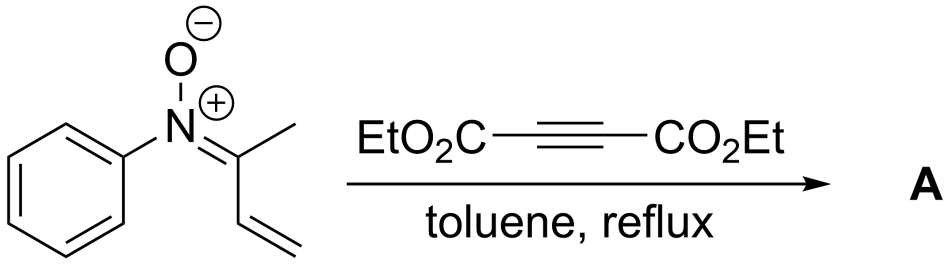
题目

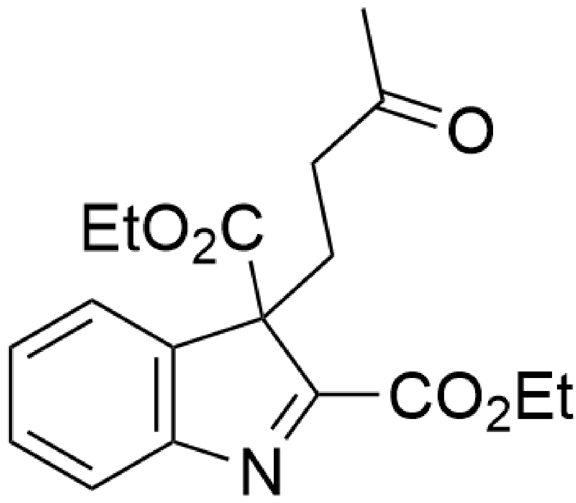


[O-]/[N+](C1=CC=CC=C1)=C(C)/C=C>O=C(C#CC(OCC)=O)OCC,toluene,reflux>[$\mathbf{A}$],$\mathbf{A}$是反应产物

人们认为，底物与丁炔二酸二乙酯反应生成含七元环的中间体$\mathbf{X}$； $\mathbf{X}$在体系的痕量水作用下转化为最终产物。已知反应产物$\mathbf{A}$中含有2个环，且$\mathbf{A}$的分子式$C_{18}H_{21}NO_5$，试给出$\mathbf{A}$的结构式

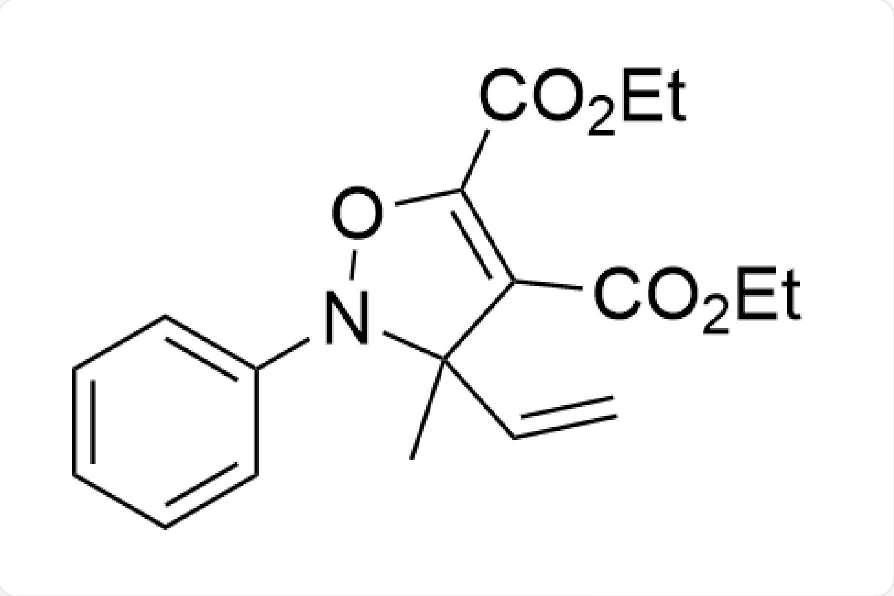
A. 其他选项均不正确

B.



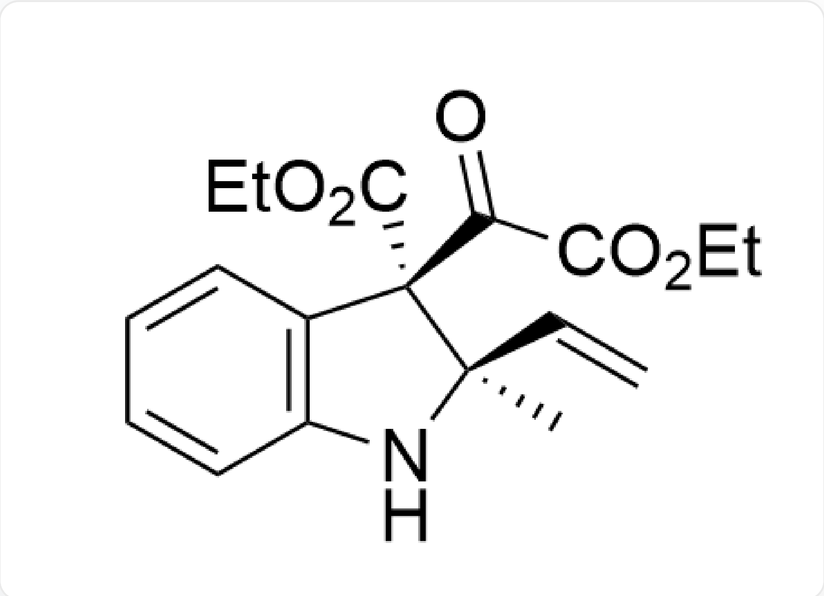
O=C(C)CCC1(C(OCC)=O)C2=C(N=C1C(OCC)=O)C=CC=C2

C.



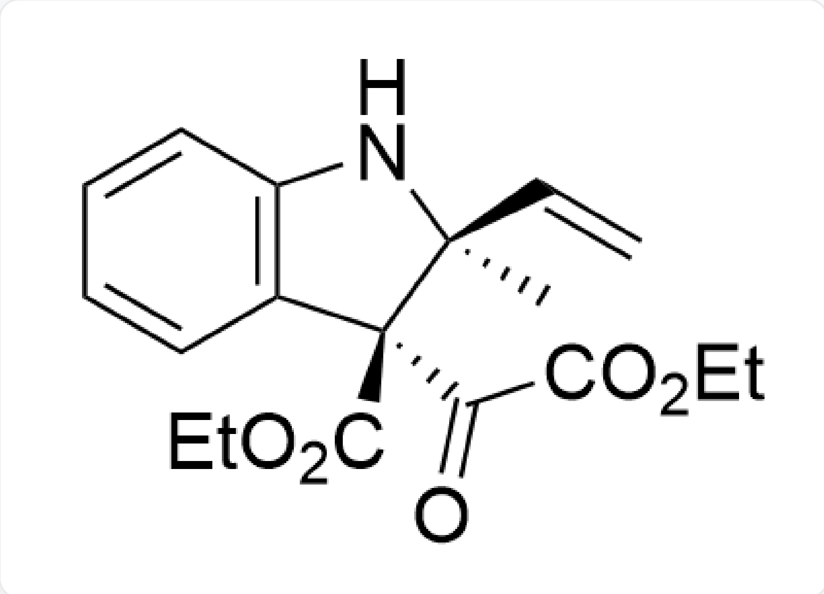
CC1(C=C)N(OC(C(OCC)=O)=C1C(OCC)=O)C2=CC=CC=C2

D.



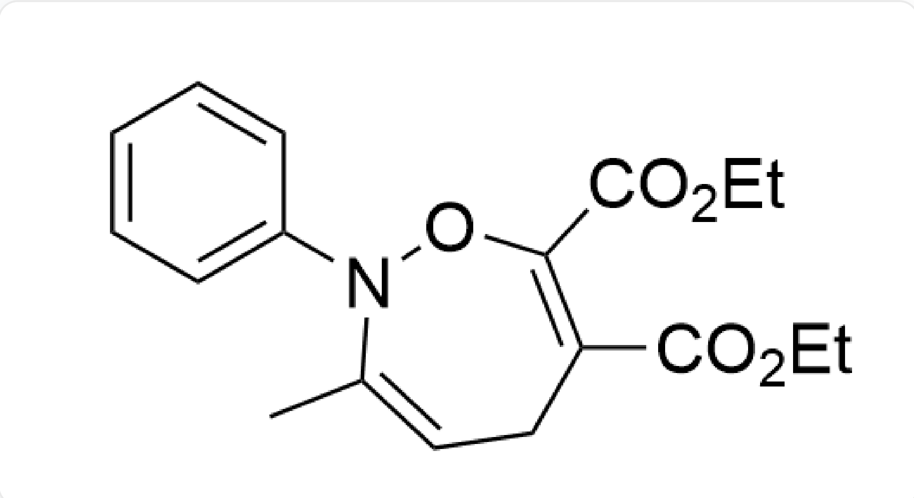
O=C(C(OCC)=O)[C@@]1(C(OCC)=O)[C@](C=C)(C)NC2=C1C=CC=C2

E.



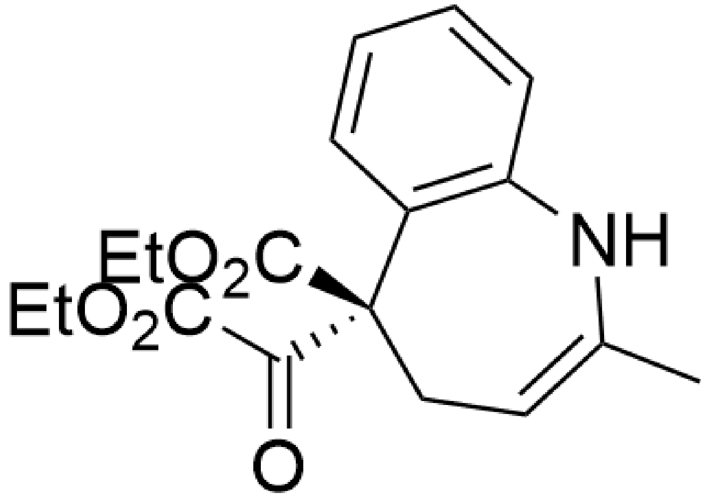
O=C(C(OCC)=O)[C@@]1(C(OCC)=O)[C@@](C=C)(C)NC2=C1C=CC=C2

F.



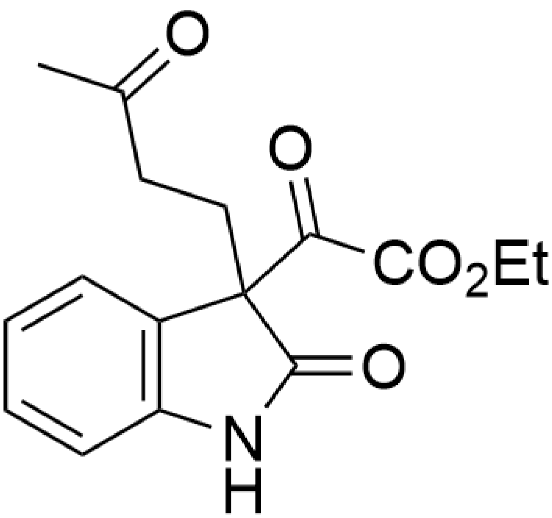
CC1=CCC(C(OCC)=O)=C(C(OCC)=O)ON1C2=CC=CC=C2

G.



O=C([C@@]1(C2=C(NC(C)=CC1)C=CC=C2)C(OCC)=O)C(OCC)=O

H.



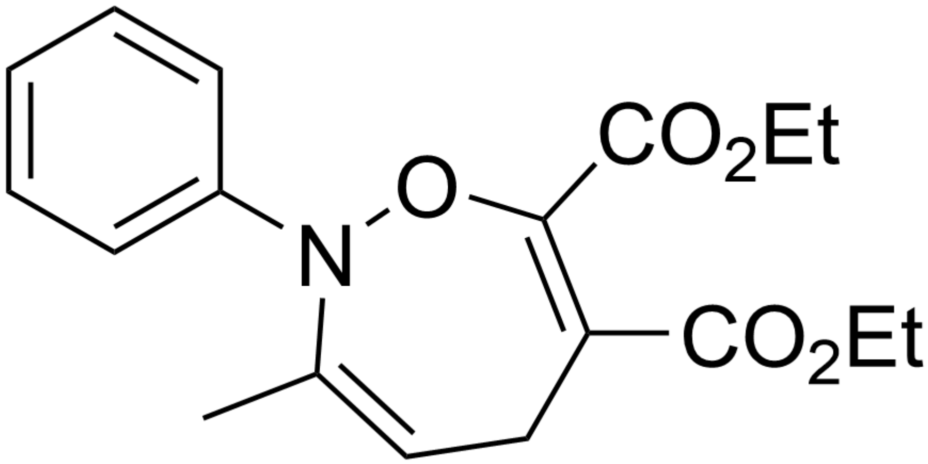
O=C(C(OCC)=O)C1(CCC(C)=O)C2=C(NC1=O)C=CC=C2

答案

正确答案: **B**

详细解析

首先根据题目提示，反应底物与丁炔二酸二乙酯反应生成含七元环的中间体 **X**

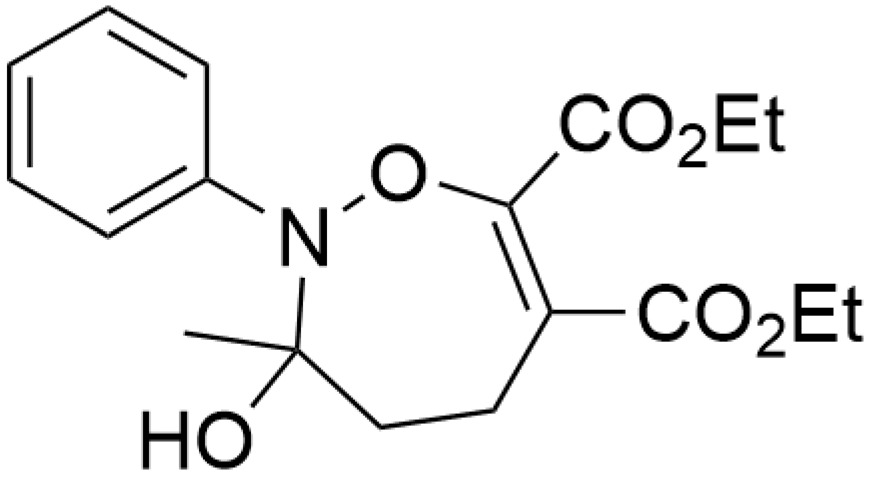


中间体**X**：CC1=CCC(C(OCC)=O)=C(C(OCC)=O)ON1C2=CC=CC=C2

CHECKPOINT 1 PTS

中间体 **X**： CC1=CCC(C(OCC)=O)=C(C(OCC)=O)ON1C2=CC=CC=C2

接着中间体 **X** 在痕量水的作用下，首先形成中间体 **1**

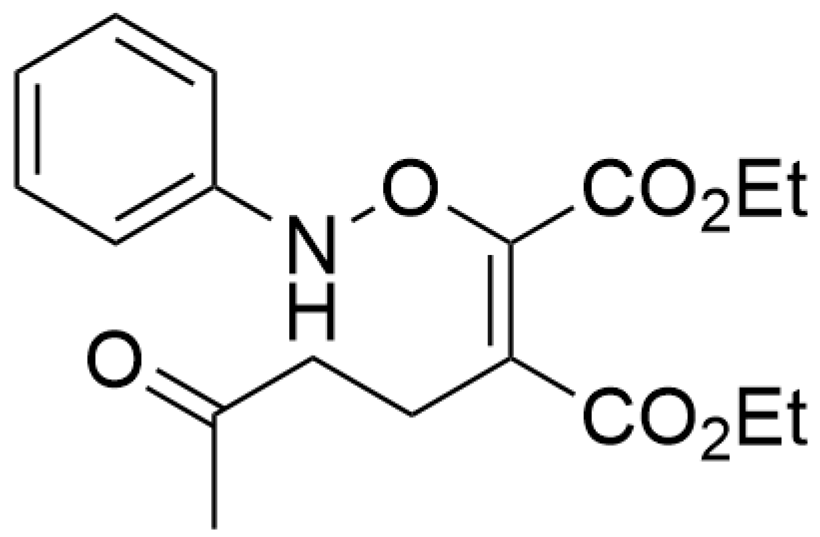


中间体**1**：CC1(O)N(OC(C(OCC)=O)=C(CC1)C(OCC)=O)C2=CC=CC=C2

CHECKPOINT 1 PTS

中间体**1**：CC1(O)N(OC(C(OCC)=O)=C(CC1)C(OCC)=O)C2=CC=CC=C2

接着中间体**1**进一步发生开环得到中间体**2**

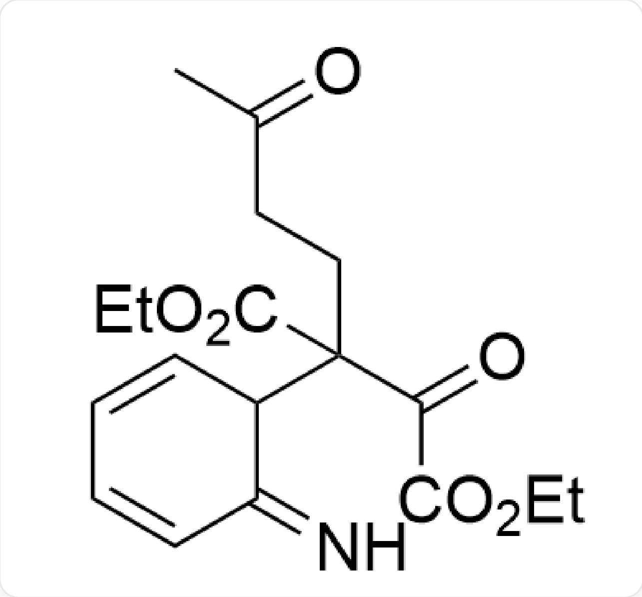


中间体**2**：CC(CC/C(C(OCC)=O)=C(C(OCC)=O)\ONC1=CC=CC=C1)=O

CHECKPOINT 1 PTS

中间体**2**： CC(CC/C(C(OCC)=O)=C(C(OCC)=O)\ONC1=CC=CC=C1)=O

此时中间体**2**的构象允许其发生3,3-σ重排反应得到中间体**3**

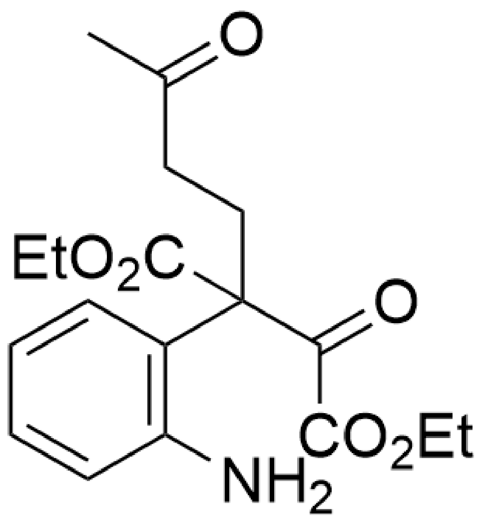


中间体**3**：N=C1C=CC=CC1C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O

CHECKPOINT 1 PTS

中间体**3**：N=C1C=CC=CC1C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O

中间体**3**发生芳构化得到中间体**4**

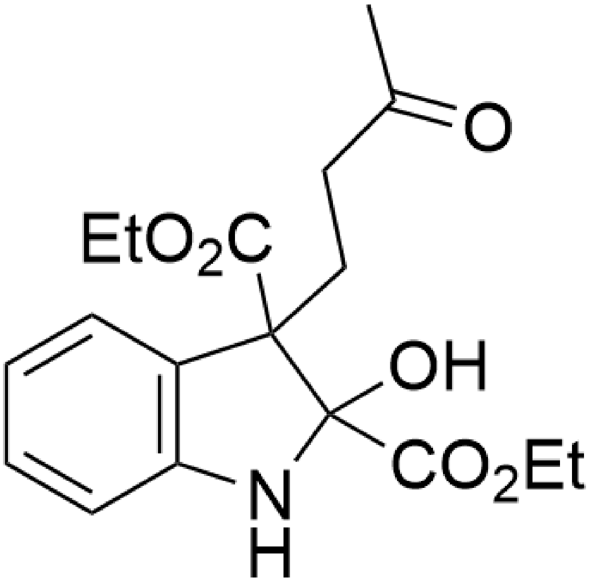


中间体**4**：NC1=C(C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O)C=CC=C1

CHECKPOINT 1 PTS

中间体**4**：NC1=C(C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O)C=CC=C1

中间体**4**中羰基的反应活性较强，因此氨基进攻羰基形成五元环，得到中间体**5**

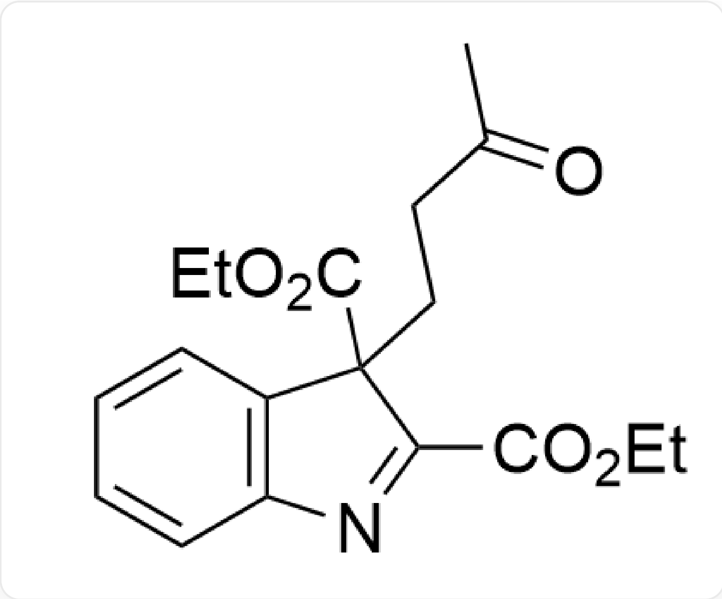


中间体**5**：OC1(C(OCC)=O)C(CCC(C)=O)(C(OCC)=O)C2=C(N1)C=CC=C2

CHECKPOINT 1 PTS

中间体**5**：OC1(C(OCC)=O)C(CCC(C)=O)(C(OCC)=O)C2=C(N1)C=CC=C2

中间体**5**继续脱去一分子水得到最终产物**A**



产物**A**：O=C(C)CCC1(C(OCC)=O)C2=C(N=C1C(OCC)=O)C=CC=C2